

CONFIDENTIAL

Indoor unit prod.date : 17-Jan-2024 Line : T
 -Lot size : (4000001 - 4000005) 5 Units
 Outdoor unit prod.date : 17-Jan-2024 Line : C
 -Lot size : (4000001 - 4000005) 5 Units
 Period of test : 22 ~ 31 January 2024
 Reference design specification no. : MCR

APPROVE (QA Dept Mgr.)	APPROVE (Dev Test SM)	CHECK	REPORT
REPORT NO.	SAM-24-760-0001		

Test result:

Classification of Evaluate Item

TestItemDetail

Outline of Evaluate Condition and Principal Standard Value

QtyJudg

Development Test result (Final)

QtyJudg

Pilot Sampling Test result

Ave (n=2)

Cooling capacity (Rate)

Room temp.

Spec: t = 50 Hz, 220 V

Tolerance

In DB27.0 °C WB19.0 °C

Q = 912 Btu/hr

Q = 912 W

Out DB35.0 °C WB24.0 °C

P = 720 W

P = 110 W

EER = 12.80

EER ≥ 9.52

SEER = 13.65 Btu/h·°F

≥ 13.17 and ≥ 97.2 % of label

EGAT Energy rating = No.5

Extension pipe length = 7.5 m

1

OK

Ave n=2 MCP (RA-8 RA-11)

Capacity = 354.0 Btu/h (101.541%)

Power Input = 714.2 W (99.154%)

EER = 13.10 (102.344%)

SEER = 13.91 (101.905%)

EGAT Energy rating = 5 *

Extension pipe length = 7.5 m

2

OK

S/N : 4000002 / 4000004 (RA-17)

Capacity = 820.8 Btu/h (103.437%)

Power Input = 722.6 W (103.361%)

EER = 13.19 (103.047%)

SEER = 13.99 (102.491%)

EGAT Energy rating = No.5 *

Extension pipe length = 7.5 m

S/N : 4000001 / 4000005 (RA-17)

Capacity = 820.6 Btu/h (102.852%)

Power Input = 715.6 W (99.389%)

EER = 12.98 (101.409%)

SEER = 13.78 (100.952%)

EGAT Energy rating = No.5 *

Extension pipe length = 7.5 m

Capacity = 840.6 Btu/h (102.145%)

Power Input = 719.1 W (99.875%)

EER = 13.09 (102.269%)

SEER = 13.89 (101.759%)

EGAT Energy rating = No.5 *

Cooling Overheat

In32.0/23.0 Out 43.0 °C

During 2 hr, protector does not work and Pst-c 4.17 Mpa

1

OK

Max. Pst = 3.402 MPa (198V)

1

OK

Max Pst = 3.04 MPa (198 V) | Max Tst = 90.1 °C (198 V)

Cooling Freezing (Dis)

In31.0/15.0 Out31.0/15.0 °C

6Hr. No ice & water drop from air outlet

1

OK

No problem without the remain ice and water drop. (0/21.0/15.0, 0/21.0/15.0 DSH = 25.3 deg)

1

OK

Have not ice on cooler surface and have not water blow out from nozzle.

Cooling Freezing (Drip)

In31.0/15.0 Out31.0/15.0 °C

8Hr. No ice accumulate & water blow out from nozzle

1

OK

No problem without the remain ice and water drop

1

OK

Have not ice on cooler surface and have not water blow out from nozzle.

Heating Overheat

In27.0 °C Out34.0/18.0 °C

During 2 hr, protector does not work and Pst-c 4.17 Mpa

Heating Freezing

Heating temp capacity

In32.0/23.0 Out 43.0 °C

Q ≥ 97.4% , P ≤ 105.6%

Automatic defrost

2 °C RH95% and ice remain 10% of Outdoor H/E

In DB20.0 °C WB12.0 °C

Time of defrosting must not exceed 20% for heating mode and Indoor outlet air

Enclosure

UIS condition

In DB21.0 °C WB1.0 °C

is lower 18°C, compressor must be not operate 1 min or more

UIS condition

RH78%, 4hr. No water drop from air outlet

Swapsitride

RH78%, 48hr.ON, 12hr.OFF, 24hr.ON

Evaporating

Refrigerant -50g 4hr. No water drop

1

OK

Evaporating 4 hr. R32-32g. No water drop (S.H / Low) (240 V).

Clogged Filter

Gauze (air filter 3, air cleaning filter 6) 4 Hr.

In32.0/23.0 Out 43.0/18.0 °C

Indoor RH78%, operated 4hr. No water drop from air outlet

Starting test

In32.0/23.0 Out 43.0/18.0 °C

Starting voltage minimum 217.5 V Starting current max -A

1

OK

Starting voltage = 168 volt (76.35%)

1

OK

Starting voltage = 167 V (75.91 %)

Low Temp. Start

Indoor temp. -5°C, outdoor temp. -10°C

Noise

UIS condition

Indoor 44 dBA outdoor 51 dBA Not over than+4.0dBA

2

OK

2

OK

Ref. Noise

NO abnormal noise

Defrost Noise

NO abnormal noise

Discharge

NO abnormal noise

Whistle Noise

NO abnormal noise

Motor Noise

NO abnormal noise

Crack Noise

No loud noise, middle level < 2time, small level < 10time

Moving Part

NO abnormal noise

pipiny noise

NO abnormal noise

Low Freq. Noise

Q/A-max,peak < 6dBA and Q/A-peak > 4dBA

Humming noise

NO abnormal noise

Feeling check

NO abnormal noise

2

OK

2

OK

Feeling check pipe 5.0 m at condition STD

Indoor start cool mode fan low have refrigerant noise smooth and compressor noise level low.

Indoor stop cool mode have refrigerant noise and compressor noise level low.

Outdoor start and stop cool mode have refrigerant noise and compressor noise level medium.

1

OK

- Feeling check pipe 3.0 m at condition STD / Freezing / Overload

- Indoor and Outdoor cool mode no have abnormal noise.

1

OK

- Feeling check pipe 3.0 m at condition Standard

- Indoor start cool mode have refrigerant noise and compressor noise level low

- Indoor running cool mode have refrigerant noise and compressor noise level low

- Indoor stop cool mode have refrigerant noise and compressor noise level low

- Outdoor running cool mode have refrigerant noise and compressor noise level low

- Feeling check pipe 3.0 m at condition Overload

- Indoor start cool mode have refrigerant noise and compressor noise level low

- Indoor running cool mode have refrigerant noise and compressor noise level low

- Indoor stop cool mode have refrigerant noise and compressor noise level low

- Outdoor running cool mode have refrigerant noise and compressor noise level low

- Feeling check pipe 3.0 m at condition Freezing

- Indoor start cool mode have refrigerant noise and compressor noise level low

- Indoor running cool mode have refrigerant noise and compressor noise level low

- Indoor stop cool mode have refrigerant noise and compressor noise level low

- Outdoor running cool mode have refrigerant noise and compressor noise level low

< ANECHOIC - 2 >

< RA -15 >

Feeling 5.0 m. by :

Mr.Pawissanul S.

Mr.Amont N.

Mr.Apiwat L.

Mr.Chayapatt S.

Vibration

Indoor Vibration

Unbalance 150, 200 μ, no vibration noise

Outdoor vibration

Vibration spec<150μm(+20 μm) not abnormally vibration

Function

Receive of signal remote control

The straight of front > 7 m and Inverter type > 4 m

Operating inspection

Operate check confirm by IB

EREC

Impulse noise

Function operate are not abnormal When voltage supply ±1.8 KV

50C / 3000 times , Play : 5000 times ON / OFF

Between lead : 3KV , Earth : 9KV for 30 second 3 times

When less than 16.5 ms, no misoperation

When more than 10 KV, Obstacle operation

-5 MHz : 56dB, 5 ~ 30 MHz : 60dB

Obstacle of electrical wave (R / E / P / E)

Not have effects to emission noise 1 m. from TV or Radio

Normal mode 40uA, Common mode 80u do not have fire or smoke

Lightning surge

Insu. resist. 1MΩ, V power sup. 1500V * 1 min

10 MΩ or more

Dielectric Withstand Voltage

Basic Insulation 1000 V / 1 min

Supplementary Insulation 1750 V / 1 min

Reinforced Insulation 3000 V / 1 min

25A between earthing and metal parts resistance not exceeding 0.1 Ω

1

OK

Maximum of Leakage Current 0.28 mA

1

OK

Max Peak : 0.29 mA, Cooler (L-E) Stop

Safety

Earthling continuity

Not exceed 2 mA per kW of max power input (≤ 1.83 mA)

The values shall not exceed the safety standard.

1

OK

Maximum of Leakage Current 0.28 mA

1

OK

Max Peak : 0.29 mA, Cooler (L-E) Stop

Structure

Moving parts

Area which hand can touch, Be have proud

Wiring

Insulation do not rupture and not attach H temperature area

Power supply cord

No tension strength at connection point

Earth terminal

Be provide separate from other terminal

Fuse

Indicate rating value can not delete

H / E / in

There is not folding or broken of fin

Easy of parts

No have gap after easy parts

Handling test

When lift the handle no damage

Band strength

Joint (welding point) 980 N or more

Life test (Reliability of part)

No damage in standard operating time (2000 ~ 4000hrs)

Heat shock

0 ~ 65°C, 1hr. 10 cycle, no have abnormal and no damage part

High temperature stock

55 °C 72hrs Plastic appearance is changing ≤ 1mm.

Endurance Ability

Packing

Accessories, IB, IM, Guarantee must be packing and check

Indication

Can not remove indication by simple way

Subject of caution

Caution must be write in IB, IM

Accessory and Caution

2. Main problem

[illegible]

		Maximum EGAT 5 star					
		NEW EGAT LEVEL					
FIXED SPEED	Capacity (kW)	5	5*	5**	5***	5****	5*****
CURRENT	≤ 8.0 kW	12.85	13.85	14.85	15.85		
	> 8.0 - 12.0 kW	12.40	13.40	14.40	15.40		
NEW EGAT	≤ 8.0 kW	13.17	13.71	14.24	14.78	15.31	15.85
(Official)	> 8.0 - 12.0 kW	12.56	13.13	13.70	14.26	14.83	15.40